

APPROVED METHOD:
PHOTOMETRY OF
REFLECTOR TYPE LAMPS
AN AMERICAN NATIONAL STANDARD

ANSI/IES LM-20-20

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AN AMERICAN NATIONAL STANDARD**

Publication of this Committee
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Suggestions for revisions
should be directed to IES.

**Prepared by:
The IES Testing Procedures Committee**



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Foreword

This approved method is a revision of *IES LM-20-13, IES Approved Method: Photometric Measuring and Reporting Tests on Reflector-Type Lamps*, published in 2013.

1.0 Introduction and Scope

1.1 Introduction

As used in this laboratory method, a *reflector type lamp* is a lamp having a reflective element(s) intended to redirect flux from the emitting element (e.g., filament, arc, LED) to form the intended spatial distribution of the light. For example, a reflective element might take the form of a reflective coating applied to the lamp bulb, or a reflector positioned relative to the emitting element and permanently affixed in this position. In addition, integrated LED lamps using reflective and refractive elements to create the intended spatial distribution are within the scope of this LM. Using the current IES definitions these lamps are called projector lamps, floodlights, pressed-reflector lamps, and spotlights, among other terms. Through popular usage, the scope of this LM covers the “bulb,” which is component of a luminaire.

1.2 Scope

This Approved Method describes photometric testing procedures and reporting guidelines for reflector type lamps and LED based lamps that mimic this traditional style of lamp. The application of the described procedures and guidelines will improve reproducibility within a laboratory, and will improve measurement agreement and facilitate comparison of results between laboratories. This laboratory method does not apply to the following: lamps of standard bulb shape to which an integral reflector is added, such as silver-bowl and silvered-neck lamps; reflector type lamps that are designed for special applications, such as automotive headlamps and projection lamps, for which lamp specific test procedures have been established; or lamps that are known to have special testing requirements beyond those addressed in this laboratory method, such as linear fluorescent reflector lamps that have special temperature or orientation requirements.

2.0 Normative References

2.1 ANSI/IES LM-35-20

Illuminating Engineering Society. Approved Method: Photometric Testing of Floodlights Using High Intensity Discharge or Incandescent Filament Lamps. New York: IES; 2020.

2.2 ANSI/IES LM-45-20

Illuminating Engineering Society. Approved Method: Electrical and Photometric Measurement of General Service Incandescent Filament Lamps. New York: IES; 2020.

For measurements of general service incandescent filament lamps, the laboratory shall meet the requirements stated therein.

2.3 ANSI/IES LM-51-20

Illuminating Engineering Society. Approved Method: The Electrical and Photometric Measurements of High Intensity Discharge Lamps. New York: IES; 2020.

For measurements of high intensity discharge lamps, the laboratory shall meet the requirements stated therein.

2.4 ANSI/IES LM-54-20

Illuminating Engineering Society. Approved Method: IES Guide to Lamp Seasoning. New York: IES; 2020.

2.5 ANSI/IES LM-63-19

Illuminating Engineering Society. IES Standard File Format for Electronic Transfer of Photometric Data and Related Information. New York: IES; 2019.

2.6 ANSI/IES LM-66-20

Illuminating Engineering Society. Approved Method: Electrical and Photometric Measurements of Single-Ended Compact Fluorescent Lamps. New York: IES; 2020.

For measurements of single-based fluorescent lamps, the laboratory shall meet the requirements stated therein.

2.7 ANSI/IES LM-75-19

Illuminating Engineering Society. Approved Method: IES Guide to Goniometer Measurements and Types, and Photometric Coordinate Systems. New York: IES; 2019.

For measurements using a goniometer system, the laboratory shall meet the requirements stated therein.

2.8 ANSI/IES LM-78-20

Illuminating Engineering Society. Approved Method: Total Luminous Flux Measurement of Lamps Using an Integrating Sphere Photometer. New York: IES; 2020.

For measurements using an integrating sphere system, the laboratory shall meet the requirements stated therein.

2.9 ANSI/IES LM-79-19

Illuminating Engineering Society. Approved Method: Electrical and Photometric Measurement of Solid-State Lighting Products. New York: IES; 2019.

For measurements of solid-state lighting products, the laboratory shall meet the requirements stated therein.

2.10 ANSI/IES LS-1-20

Illuminating Engineering Society. Lighting Science: Nomenclature and Definitions for Illuminating Engineering. New York: IES; 2020. Online: www.ies.org/standards/definitions/. (Accessed 2019 Apr 17).

3.0 Definitions

Electrical measurement units are the volt, the ampere, and the watt. The unit of photometric flux is the lumen.

The unit of luminous intensity is the candela. The unit of temperature measurement is the degree Celsius.

3.1 beam axis

The beam axis for lamps designed to produce rotationally symmetrical beam patterns is defined as the line in the direction of the beam propagation about which the intensity distribution is substantially symmetric. It follows that any rotation of the projected beam pattern about the beam axis will result in a substantially similar projected beam pattern. The beam axis for quadrilateral symmetry is the intersection of the two planes about which the beam has reflected symmetry. The beam axis is established by the spatial relationship of the source of the light in the lamp and the beam-forming optical components of the lamp, such as reflective surfaces and lenses. The beam axis is not affected by the placement of non-optical lamp components, such as the lamp base, or any ancillary hardware, such as lamp holders. If the beam has no substantial symmetry about an axis in the direction of the propagation of the beam, the beam axis is undefined (see **Figure 3-1B**).¹

3.2 central cone

The photometric center of the lamp is the vertex of a cone where the directrix of the cone is defined by a constant angle from the photometric axis, n° . This is referred to as a “ $2n^\circ$ central cone” (see **Figure 3-2**). Typically, lamp design is such that the photometric axis and the beam axis are intended to coincide. By definition, the photometric axis contains the photometric center; therefore the beam axis also includes the photometric center, by intent if not in practice.

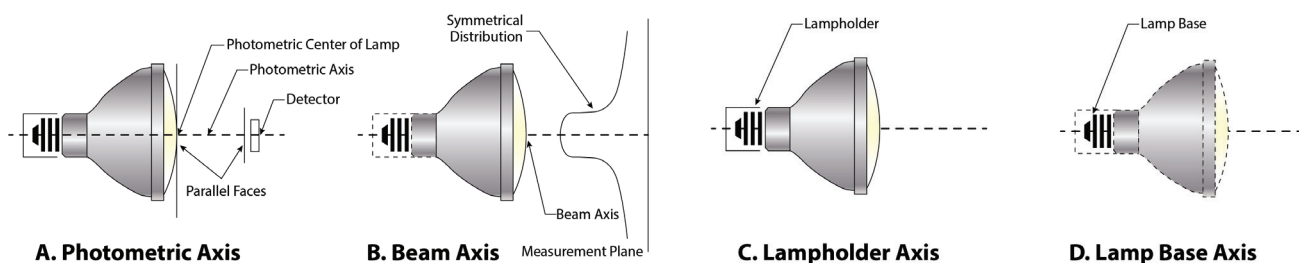


Figure 3-1A-D: Beam axis types. (© Illuminating Engineering Society)

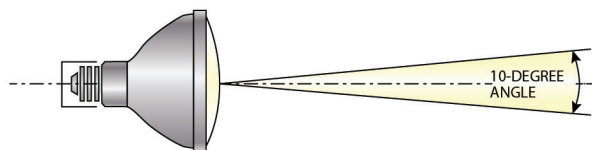


Figure 3-2. Example of a central cone where $n=5^\circ$.

(© Illuminating Engineering Society)

3.3 extraneous light

Light from sources other than the lamp being tested that passes through the complete optical system to the detector along the intended light path. Extraneous light will have associated with it a stray-light component. Extraneous light can be only be eliminated by eliminating the source of the extraneous light itself.

3.4 photometric axis

The photometric axis is the line through the photometric center of the lamp and the center of the detector aperture (see **Figure 3-1A**).

3.5 photometric center of lamp

The location of the photometric center of a lamp is often difficult to determine and may be subject to interpretation (see **Figure 3-1A**). The photometric center of the lamp is mounted at the intersection of the goniophotometer axes.

3.5.1 For Non-symmetric Beam Patterns. For lamps with beam patterns that exhibit no symmetry about the direction of the beam propagation, the photometric center of the lamp is the geometric center of the physical aperture of the lamp that limits the beam. This aperture may be the edge of the reflectorized surface where the beam leaves the lamp, or of the lens if a lens is used. Protuberances or recesses on the face of the lens that are present by design shall not be used in the determination of the location of the photometric center.

3.5.2 For Symmetric Beam Patterns. For lamps with beam patterns that exhibit symmetry about the direction of the beam propagation, the photometric center of the lamp is the point of intersection of the beam axis as defined in **Section 3.4 photometric axis** and a plane through the edge of the physical aperture

of the lamp that limits the beam. This aperture may be the edge of the reflectorized surface where the beam leaves the lamp, or of the lens if a lens is used. Protuberances or recesses on the face of the lens that are present by design shall not be used in the determination of the location of the photometric center.

3.6 stray light

Light from any source, including the lamp being tested, that does not pass as intended through the complete optical system before reaching the detector. Stray light is due to light leaks into the optical path, reflection and scattering from surfaces and optical mounts, reflection and scattering at surfaces of optics, and scattering or fluorescence from within the optics. Stray light can be minimized through baffling, component selection, reducing the number of reflective surfaces, and control of the reflectance of surfaces.

3.7 undirected light

Light incident upon the reflector that is not directed by reflection into the beam of light delivered by the lamp; this can be intentionally, such as in the case of wavelength selective reflection, or unintentionally, such as through loss due to imperfections in the reflector or uncoated regions in the neck of the lamp.

4.0 Ambient and Physical Conditions

4.1 General

Provisions shall be made for correct optical positioning of the lamp in relation to the photometric axis and for intensity measurements at any required angular setting.

4.2 Temperature

Requirements are provided in ANSI/IES LM-45-20 for incandescent (including tungsten-halogen), ANSI/IES LM-51-20 for high intensity discharge (HID), ANSI/IES LM-66-20 for compact fluorescent (CFL), and ANSI/IES LM-79-19 for light emitting diode (LED) light sources (see **Section 2**). For heat-sensitive lamps, such as compact

fluorescent reflector lamps (CFLr) for which the ambient temperature required is $25^{\circ}\text{C} \pm 1^{\circ}\text{C}$, care shall be taken to avoid increasing the temperature of the lamp as a result of its being inside the simulated luminaire during stabilization. One proven method to accomplish this is to first burn the lamp in open air below the luminaire until the lamp is photometrically stable, and then lower a simulated luminaire over the lamp and immediately take a measurement.

4.3 Airflow

Requirements are provided in ANSI/IES LM-45-20 for incandescent (including tungsten-halogen), ANSI/IES LM-51-20 for high intensity discharge (HID), ANSI/IES LM-66-20 for compact fluorescent (CFL), and ANSI/IES LM-79-19 for light emitting diode (LED) light sources (see **Section 2**).

4.4 Vibration

Lamps should not be subjected to excessive vibration or shock during measurement.

5.0 Electrical and Photometric Test Conditions

5.1 Wave Shape

Requirements are provided in ANSI/IES LM-45-20 for incandescent (including tungsten-halogen), ANSI/IES LM-51-20 for high intensity discharge (HID), ANSI/IES LM-66-20 for compact fluorescent (CFL), and ANSI/IES LM-79-19 for light emitting diode (LED) light sources (see **Section 2**).

5.2 Voltage or Current Regulation

Requirements are provided in ANSI/IES LM-45-20 for incandescent (including tungsten-halogen), ANSI/IES LM-51-20 for high intensity discharge (HID), ANSI/IES LM-66-20 for compact fluorescent (CFL), and ANSI/IES LM-79-19 for light emitting diode (LED) light sources (see **Section 2**).

5.3 Circuits and Lamp Connections

Requirements are provided in ANSI/IES LM-45-20 for incandescent (including tungsten-halogen), ANSI/IES LM-51-20 for high intensity discharge (HID), ANSI/IES

LM-66-20 for compact fluorescent (CFL), and ANSI/IES LM-79-19 for light emitting diode (LED) light sources (see **Section 2**).

5.4 Photometer System

Provisions shall be made for correct optical positioning of the lamp in relation to the photometric axis and for intensity measurements at any required angular setting. The photometer system, including detector, amplifier, and readout device, shall be evaluated for error as indicated in CIE 69-1987² specifically, but not limited to f_1' (photopic correction), f_3 (detector non-linearity), f_4 (readout error), and f_5 (detector fatigue). It is recommended that wherever possible, detectors with an f_1' value smaller than 1.5 should be used.

It is possible to modify the detector response curve to correct for the spectral non-linearity of system response due to the use of mirror and aperture glass. A detector would have to be tailor-made to the specific system, but as a result the entire system (e.g., detector, mirror, aperture included) would have an f_1' value that is smaller than 1.5.

Instrument performance shall be adequate for the level of uncertainty claimed for any reported data or derived values. Stray light shall be minimized, and where possible the residual stray light response should be subtracted from the photometric test data. All possible efforts to characterize and eliminate extraneous light shall be implemented. Residual electronic offsets of the photometric system (e.g., dark noise) shall be characterized and subtracted from the photometric test data or shown to be insignificant.

6.0 Lamp Test Procedures

6.1 Lamp Seasoning

Test lamps shall be seasoned appropriately to provide the necessary stabilization for photometric measurements. ANSI/IES LM-54-20 provides additional information (see **Section 2.4**).

6.2 Lamp Stabilization

During the measurement itself, lamps shall be stabilized before the readings are taken. Requirements are provided in ANSI/IES LM-45-20 for incandescent (including tungsten-halogen), ANSI/IES LM-51-20 for high intensity discharge (HID), ANSI/IES LM-66-20 for compact fluorescent (CFL), and ANSI/IES LM-79-19 for light emitting diode (LED) light sources (see **Section 2**).

7.0 Photometric Characterization by Measurement of Intensity Distribution

7.1 Goniophotometer

Intensity distribution measurements of a lamp are made at positions on the surface of an imaginary sphere, with the photometric center of the lamp at the center of revolution of the goniophotometer. Descriptions of the types of goniophotometers can be found in ANSI/IES LM-75-19, which describes the terms used in identifying the positioning angles of the goniometer (see **Section 2.7**). ANSI/IES LM-75-19 also describes the spherical coordinate system that is used to identify the angular direction of data points. In addition, the discussion of goniophotometers and/or lamp handling methods in ANSI/IES LM-11-20,³ ANSI/IES LM-35-20 (see **Section 2.1**), and ANSI/IES LS-4-20, *Lighting Science: Measurement of Light – The Science of Photometry*⁴ are recommended for additional details. It is strongly recommended for incandescent but required for HID, linear fluorescent, and compact fluorescent lamps to keep the lamp fixed in its normal operating orientation unless the purpose of testing dictates a different orientation; it is preferred that the lamp be kept stationary. This eliminates changes in the lamp output caused by motion or by a change in the orientation of the lamp relative to the local gravity gradient. This is particularly critical for HID lamps, as the total flux and the intensity distribution are affected by the location of the metal halide condensate, and even movement of the lamp in line with the direction of gravity can cause displacement of the liquid, can cause mobile additive condensate, and can alter the lamp operation.

Type A and Type B goniophotometers change the orientation of the lamp with respect to the direction of gravity during measurement; therefore, they are not recommended for measurement of the intensity distributions of saturated discharge lamps, such as HID and fluorescent lamps.

Given the potential effect of changes in lamp orientation, a Type C goniophotometer is more generally applicable across lamp technologies.

7.2 Goniophotometer, Optical Alignment

For lamps having a defined beam axis, the photometric axis and beam axis shall be made to coincide as closely as possible by physical alignment prior to measurement. For an ideal lamp and lamp holder mounted in a perfectly aligned goniophotometer, the beam axis, the photometric axis, and the lamp holder (and/or lamp base) axis coincide. However, variations inherent in the manufacture of reflector type lamps may cause the beam axis to differ from the axis of the lamp base. The orientation of the combination of the lamp and lamp holder should be adjustable to allow for alignment of the beam axis to the photometric axis (see **Figure 7-1**).

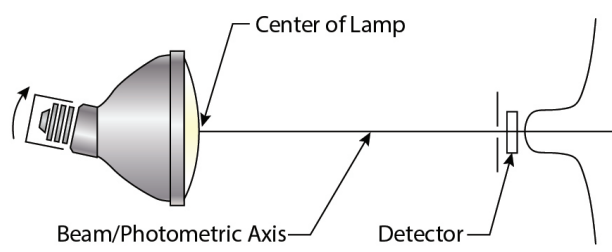


Figure 7-1. The lamp is shifted to place the beam axis parallel to the photometric axis. (© Illuminating Engineering Society)

7.3 Test Distance

The photometric center of the lamp shall be used as the point from which to measure the test distance for goniometric measurement. Therefore, for goniometric measurement, the photometric center of the lamp shall coincide with the center of revolution of the goniophotometer. Due to the wide range of lamp designs, it is required that the definition of *photometric center* used to determine the alignment of the lamp with the goniophotometer center be specified when reporting goniophotometer-based results.

For intensity distribution measurements, the distance between the lamp and the detector shall be such that the inverse-square law applies. In general, the minimum distance for the inverse-square law to be valid is the distance beyond which a constant fraction of the exit pupil of the source remains ‘flushed’ or filled. (To verify that the test distance is sufficient for far field assumptions, see ANSI/IES LM-11-20.³) The required distance varies with the effective size of the lamp. Since the optical scheme used to produce the beam of a reflector type lamp can be complex, it is easiest to determine the minimum required measurement distance by experimentally measuring the variation of intensity with distance to find the minimum distance at which the inverse-square law applies.

Illuminance levels at the detector should be optimized to reduce the uncertainty of measurement. Low illuminance levels may lead to increased uncertainty of measurement, and high illuminance levels may saturate the detection system. The illuminance levels at the detector associated with the spatial intensity distributions vary from a maximum at some point in the primary, intended beam to low illuminance levels in the regions outside of the intended beam. Therefore, the usual practice is to use the shortest test distance that is compatible with the inverse-square law and that provides for sufficient headroom for the detection system, based on the maximum anticipated illuminance.

If the detection system gain is fixed during measurement of the complete distribution, the illuminance outside of the beam or even at the extremes of the beam may not be in the optimal detection range. For the purposes of beam characterization, this is usually not a significant issue. However, an example of where this typical guidance on illuminance level would not be the optimal solution would be where the characteristics of the spatial intensity distribution outside of the intended beam are the information of primary interest. In such a case, it might be best to select a shorter test distance that still meets the requirements of the inverse-square law, which results in the maximum illuminance saturating the detection system but optimizes the measurement of the lower illuminances of interest. If the detection system is capable of automatically adjusting the gain to optimize the detection system during beam characterization, this is of less concern.

7.4 Angular Resolution

The angular resolution required for accurate characterization of the test lamp intensity distribution depends on the symmetry of and the intensity gradients present in the beam. For an irregularly shaped beam or where the beam intensity is changing rapidly with respect to direction, a finer angular grid will be needed.

Table 7-1 shows the recommended angular resolution based on the beam angle of the lamp. Using a finer resolution than recommended is acceptable and may even be beneficial.

Table 7-1. Recommended Angular Resolution (Degrees) for Various Beam Angles

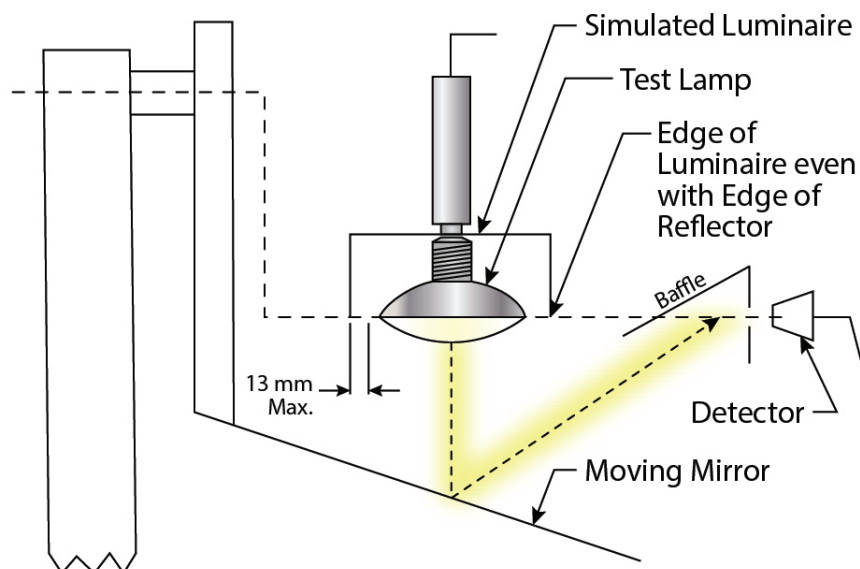
For Lamps with Beam Angles < 20°	
% of MBCP*	Angular Resolution (Degrees)
100% - 50%	1
<50% to 10%	2
< 10%	5
For Lamps with Beam Angles ≥ 20°	
% of MBCP*	Angular Resolution (Degrees)
100% to 50%	2
< 50%	5

* MBCP = maximum beam candlepower

7.5 Undirected Light

Undirected light shall be excluded from the measurement. This may be accomplished by post-processing to remove the contribution of undirected light, or by use of a simulated luminaire to prevent undirected light from reaching the detector. With the latter method, the test lamp is placed in a simulated luminaire (see **Figure 7-2**).

A typical luminaire is cylindrical with one open end. The closed end of the cylinder shall have a suitable lamp holder or other hardware to support and permit electrical connection of the lamp. The length of the lamp holder shall be such that the boundary of the beam-limiting aperture of the test lamp is aligned with the edge of the cylinder opening. The bulb-to-luminaire clearance at the opening of the luminaire shall not exceed 13 mm (0.5 in.). The luminaire shall



Note: Luminaire painted flat black inside and outside

Figure 7-2. Use of a simulated luminaire in goniophotometry. (© Illuminating Engineering Society)

be designed to prevent reflected light from the inner luminaire surface from reaching the photometer. This is sufficiently accomplished by painting the luminaire flat black inside and outside.

For some lamps, undirected light can also be eliminated from the measurement results during post-processing by limiting the angular region over which the zonal integration of the light flux is calculated. For example, limiting the angular region over which the zonal lumens are calculated for total flux determination from 0° (beam axis) to $+90^\circ$ or to -90° is sufficient to eliminate the undirected light for some sources.

Many laboratories use nothing but post-processing as the only tool to correct for undirected light. Unfortunately, post-processing ignores the fact that the side (neck and reflector body) of a bare lamp is in direct view of the detector (or mirror) once it moves out from under the lamp. Thus, the light that is supposed to be blocked by the simulated luminaire will get mixed with the down-light. This error cannot be corrected by post-processing.

An acceptable alternative to a full simulated luminaire is the use of a collar (tube) that is open on both ends. This

collar can be placed on the lamp in situations when the side light needs to be blocked from the detector (or mirror). The height of the collar has to be sufficiently long to cover the full length of the lamp. It can be as simple as rolled-up black stock paper stapled into the right size forming a collar around the lamp.

7.6 Number of Distribution Curves

Data collected by a goniophotometer can oftentimes be presented in a simplified format, depending on the symmetry of the intensity distribution.

The intensity distribution of a lamp whose beam is nominally circular in cross section may be represented by a single intensity distribution curve in the horizontal plane. The intensity distribution of a lamp whose beam is nominally oval or rectangular in cross section shall be represented by at least two intensity distribution curves, at the angles aligned with the major and minor axes of the distribution. The need for intensity distribution curves at additional angles and the angular increments required are dependent on the expected light intensity distribution (symmetry of the beam) from the lamp and the purpose of the measurement.

7.7 Polar Goniophotometer Based Measurement

For total luminous flux measurements of lamps, the Type C or polar goniophotometer design is generally used. The methods described in this Approved Method assume this goniophotometer geometry, except as noted.

7.8 Beam with Rotational Symmetry

A plot of the intensity in a plane containing the beam axis represents the intensity distribution of a nominally circular beam. (Note that the essentially circular nature of the beam establishes the existence of a beam axis.) There are two commonly used approaches to determining the required intensity distribution for a lamp mounted such that the beam axis is oriented along the rotational axis of the goniophotometer.

The predominant method using the Type C goniophotometer geometry, the polar scan method, determines the intensity distributions at a fixed lateral angle over a 360° range in the vertical angle by taking measurements with the detector aperture rotating along meridians about the lamp. This measurement is performed for a set of equally spaced lateral angles, with the spacing that which is determined to be necessary to sufficiently map the expected intensity distribution characteristics of the lamp. The scans at each lateral angle are then averaged by vertical angle to obtain an average intensity for each required vertical angle.

A second, less used method also using the Type C geometry, the rotation of the lamp method, determines the average intensity at a fixed vertical angle, over a 360° range in the lateral angle, by taking measurements with the lamp rotating about the beam axis. This measurement is performed for each of a set of vertical angles determined to be necessary to sufficiently map the expected intensity distribution of the lamp.

7.8.1 Polar Scan Method. Eight 180°-vertical-angle intensity distributions spaced at 45° lateral angle intervals about the beam axis are usually sufficient for assessment of lamps with symmetric intensity distributions. Smaller intervals may be required if there is limited symmetry of the beam or if the symmetry of the beam is finely structured. Any discharge lamp

operating in a saturated mode—i.e., not all of the lamp chemical dose is in the vapor state during normal operation—will change its output if its orientation with respect to the gravity gradient is altered. In addition, movement not resulting in a change in orientation relative to gravity may disturb the condensate and change the operation of the lamp. This method may be implemented such that intermittent rotation about the polar axis is required to change the lateral angle; in this case, the cautions about the effect of movement of discharge lamps without a change in orientation relative to the direction of gravity apply. (It is also possible to change lateral angle by changing the location of the detector aperture.)

7.8.2 Rotation of Lamp Method. Using a polar goniophotometer with alignment of the beam axis along the polar axis, the axis of lamp rotation is vertical. The allowed speed of lamp rotation can be verified by comparison of this method with an alternative method (see **Section 7.8.1**), or by verification with separately characterized lamps of similar design. Any discharge lamp operating in a saturated mode—i.e., not all of the lamp chemical dose is in the vapor state during normal operation—will change its output if its orientation with respect to the gravity gradient is altered. In addition, movement not resulting in a change in orientation relative to gravity may disturb the condensate and change the operation of the lamp. As this method requires continuous rotation about the polar axis, the cautions about the effect of movement of discharge lamps without a change in orientation relative to the direction of gravity apply.

7.9 Beam without Rotational Symmetry

It is possible to make these measurements with the Type C goniophotometer; typically for many incandescent lamp types, measurements on lamps of such beam design are made using Type A or Type B goniophotometers. The guidance applied for the beam with rotational-symmetry geometry (see **Sections 7.8.1** and **7.8.2**) applies with the following additional comments:

- The intensity distribution of a lamp whose beam is nominally oval or rectangular in cross section shall be represented by at least two intensity distribution curves at the azimuthal angles aligned with the major and minor axes of the distribution.

- The need for intensity distribution curves at additional azimuthal angles and the polar and azimuthal increments required are dependent on the expected intensity distribution of the lamp and the purpose of the measurement; sufficient measurements shall be made to achieve the intent of the measurement.

7.10 Determination of Derived Quantities from the Intensity Distribution

7.10.1 Luminous Flux. The total luminous flux can be calculated from an average intensity distribution curve for a lamp of nominally circular light distribution about the beam axis by using zonal lumen constants for annular zones.

The formula for calculating the zonal lumens depends on the zonal constant, K :

$$K(\phi, \theta) = \pi \cos \theta_1 - \cos \theta_2 (\phi_2 - \phi_1) / 180, \quad (7-1)$$

where θ is the polar angle and ϕ is the vertical angle, in degrees.

Evaluating with ϕ_2 at 360 and ϕ_1 at 0 degrees (i.e., around the spherical surface at a given polar angle) gives:

$$K(\theta) = 2\pi(\cos \theta_1 - \cos \theta_2) \quad (7-2)$$

To obtain the number of lumens (zonal lumens, ZL) in each polar-angle strip, the zonal constant K is multiplied by the average luminous intensity (I) of the zone.

Assuming that intensity measurements exist at even polar angles, the zonal lumens are found from:

$$ZL = K(I_1 + I_2 + \dots + I_n)/n, \quad (7-3)$$

where n is the number of intensity measurements in the given zone.

Several example sets of zonal constants are listed in **Table 7-2**.⁵

Table 7-2. Constants to Use in the Zonal Method of Computing Luminous Flux from Luminous Intensity Data

1-Degree Zones		2-Degree Zones		5-Degree Zones		10-Degree Zones	
Zone Limits (degrees)	Zonal Constant	Zone Limits (degrees)	Zonal Constant	Zone Limits (degrees)	Zonal Constant	Zone Limits (degrees)	Zonal Constant
0 – 1	0.0009	0 – 2	0.0038	0 – 5	0.0239	0 – 10	0.095
1 – 2	0.0029	2 – 4	0.0115	5 – 10	0.0715	10 – 20	0.283
2 – 3	0.0048	4 – 6	0.0191	10 – 15	0.1186	20 – 30	0.463
3 – 4	0.0067	6 – 8	0.0267	15 – 20	0.1649	30 – 40	0.628
4 – 5	0.0086	8 – 10	0.0343	20 – 25	0.2097	40 – 50	0.774
5 – 6	0.0105	10 – 12	0.0418	25 – 30	0.2531	50 – 60	0.897
6 – 7	0.0124	12 – 14	0.0493	30 – 35	0.2946	60 – 70	0.993
7 – 8	0.0143	14 – 16	0.0568	35 – 40	0.3337	70 – 80	1.058
8 – 9	0.0162	16 – 18	0.0641	40 – 45	0.3703	80 – 90	1.091
9 – 10	0.0181	18 – 20	0.0714	45 – 50	0.4041		
				50 – 55	0.4349		
				55 – 60	0.4623		
				60 – 65	0.4862		
				65 – 70	0.5064		
				70 – 75	0.5228		
				75 – 80	0.5351		
				80 – 85	0.5434		
				85 – 90	0.5476		

The zones used shall be sufficiently small that negligible error is introduced from variation in luminous intensity. For example, a 10° beam may require 1° zones, while 5° zones may be adequate for a 60° beam. (See **Table 7-1**, in **Section 7.4**, for the recommended angular resolution based on the beam angle of the lamp.) The validity of a flux determination made by these means depends upon the degree to which the intensity distribution represents the true average distribution about the lamp.

7.10.2 Beam Angle. The beam angle is designated as the total angular spread of the cone intercepting the intensity that is 50% of the maximum intensity. (Refer to ANSI/IES LS-1-20 [see **Section 2.10**] and ANSI C78.379¹ for more details.)

7.10.3 Field Angle. The field angle is designated as the total angular spread of the cone intercepting the intensity that is 10% of the maximum intensity. (Refer to ANSI/IES LS-1-20 [see **Section 2.10**] for more details.)

7.10.4 Beam Flux Values. Beam flux may be calculated from an average intensity distribution curve or from an isocandela diagram. Of particular interest is the flux contained within the limits of the beam and field angles. The calculation procedure is the same as for total flux, except that only flux within the beam angle limit or angular cone is included. For a lamp designed to have an off-axis maximum intensity, the beam and field angles are referenced to the axis of the maximum intensity.

7.10.5 Beam Pattern Classification. The classification of beam patterns is based on the beam angle. (Refer to ANSI C78.379 for beam pattern descriptions and maximum beam candlepower, i.e., luminous intensity, illustrations.¹)

7.11 Special Applications

To simplify comparison, when reflector type lamps are considered for a special application, the lamp intensity distribution results shall be presented in the same form as that specified by the applicable ANSI/IES Light Measurement (LM) standard for the equipment used for the same application. For example, when a direct comparison of a reflector type lamp with a floodlighting luminaire is needed, the lamp shall be tested according to approved floodlighting test procedures.

8.0 Total Flux Measurement by Integrating Sphere Method

Integrating sphere measurements are capable of producing total flux measurements only. All spatial information is inherently lost with this measurement method. The basics of integrating sphere photometer calibration and measurements are as described in ANSI/IES LM-78-20 (see **Section 2.8**).

8.1 Calibration of Integrating Sphere

Total flux standards that have near-spherical or directional flux distribution patterns may be used to calibrate the integrating sphere. It is desirable that standard lamps used for calibration for total flux of reflector type lamps have a beam distribution similar to that of the test lamp. When using near-spherical distribution flux standards, the extent of the difference in the illumination of the sphere walls by the standard and test sources determines the potential for systematic error due to calibration. This potential can be reduced by selection of the sphere and location of the test lamp so as to illuminate as much of the sphere wall as possible. In addition, high-reflectivity integrating sphere walls (greater than 95%) will reduce the effect of the difference in illumination by the standard and the test lamps. The testing laboratory shall document its ability to accurately apply this method of calibration to the type of lamp being tested.

8.2 Exclusion of Undirected Light by Using a Luminaire inside an Integrating Sphere

Where there is a need to exclude undirected light from the measurement, lamps shall be tested in a simulated luminaire (see **Figure 8-1**). The simulated luminaire is also described in **Section 7.5**, with the exception that the outside of the luminaire shall be coated with white photometric-sphere paint. If the luminaire is small with respect to the size of the integrating sphere (diameter approximately 10% of the sphere diameter or smaller), flat white enamel is satisfactory for this purpose. Other means of excluding undirected light may be used if they produce the same results as the recommended simulated luminaire described here.

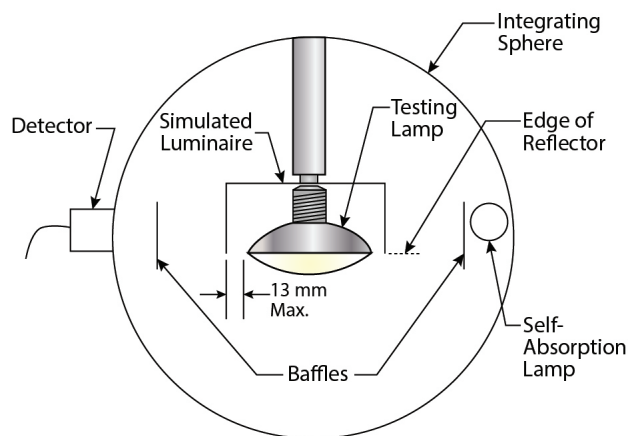


Figure 8-1. Use of a simulated luminaire in sphere photometry (luminaire is painted with flat black paint on the inside and white photometric-sphere paint on the outside). (© Illuminating Engineering Society)

8.3 Measuring Compact Fluorescent Reflector with an Integrating Sphere

Compact fluorescent reflector lamps (CFLr) have two characteristics that require special attention. First, CFLr performance is more sensitive to ambient temperature than are other light sources covered by this Approved Method. Consequently, CFLr's require a different stabilization procedure when applying the simulated-luminaire integrating sphere measurement (see **Section 4.2 Temperature**). Second, a portion of the light-emitting area of the discharge tube of a CFLr may not be completely obscured by the reflector when viewed from polar angles greater than or equal to 90°. That part of the discharge tube not obscured by the reflector will radiate omni-directionally. In one example, 8% of the total emitted light measured on a Type-C goniophotometer was measured between polar angles of 90° and 130°. This uplight is easily deducted from consideration of the effective light output of a CFLr if a goniometric measurement is made, but it is not possible to fully correct for uplight when a simulated luminaire measurement in an integrating sphere is performed. Therefore, the sphere-measured luminous flux will not be equal to the goniophotometer-measured downlight (light emitted in vertical angles less than 90°.)

The use of the simulated luminaire for reflector lamps was intended for the limited purpose of blocking the light transmitted through the neck area of the lamp or

through the reflector surface toward the base of the lamp. The simulated luminaire is not effective in blocking the omnidirectional light from a section of the discharge tube not obscured by the reflector. Light incident on the interior of the simulated luminaire (entering the gap between the reflector body and the edge of the luminaire) is mostly absorbed, but light incident on the edge of the luminaire will be partially reflected and partially absorbed, with the reflected light contributing to the measured total luminous flux. The light at polar angles greater than 90° that is not incident on the luminaire will also contribute to the measured total luminous flux.

Making a perfect simulated luminaire that will fully eliminate the uplight in the sphere created by the below-reflector portion of the CFLr light source is not possible. Measurement differences between goniometer and sphere measurements for an individual laboratory and between accredited laboratories will be larger than for reflector lamp types where the light source is completely obscured by the reflector for vertical angles greater than 90°. For interchanges conducted between competent laboratories on non-CFLr's, differences of 2% to 3% are considered within measurement acceptability, while acceptable differences for CFLr's may be 4% to 6%.

9.0 Test Report

9.1 Lamp and Test Description

The test report shall include all significant data including the measured luminous intensity distribution and the parameters pertinent to the purpose of the test. Typical items reported are covered in the subsections that follow.

9.1.1 Lamp, General.

- Manufacturer's description
- Identifying lamp number
- Name of testing laboratory
- Manufacturer or brand
- NAED code
- Beam type
- Bulb type

- Base type
- Filament type
- Rated lumen output
- Rated electrical values

9.1.2 Sphere-Specific.

- Date of test
- Measured electrical values
- Sphere size
- Luminous flux
- Color, correlated color temperature, color rendering index (or other color rendering rating; e.g., from ANSI/IES TM-30-18⁶)

9.1.3 Gonio-Specific.

- Date of test
- Measured electrical values
- Up, down and total luminous flux
- Beam angle, field angle
- Beam lumens, field lumens
- Max beam candlepower (MBCP), center-beam candlepower (CBCP)
- IES file (for more information on IES photometric file format, refer to ANSI/IES LM-63-19; see **Section 2.5**)

9.2 General Test Conditions

The report shall list all pertinent information on the general test conditions. Typical items reported are:

- Number of lamps tested
- Location of photometric center applied
- Test distance
- Orientation of lamp during test (horizontal, vertical)
- Lamp movement (rotating, stationary)
- Type of goniophotometer and coordinate system used
- Method of making sphere measurements (lamp position, type of standard) and sphere description (diameter, sphere paint reflectivity)

9.3 Lamps Designed for General Lighting

Reports on lamps for general lighting applications shall be in a form directly comparable to reports on general lighting luminaires.⁷

9.4 Lamps Designed for Spotlighting and Floodlighting

The report shall list all pertinent data on lamps for spotlighting and floodlighting. Typical items reported are:

- Total flux, in lumens
- Average intensity, in candelas, for central cone¹
- Beam angle, in degrees
- Beam flux, in lumens
- Additional information on the distribution pattern, in one of the following forms:
 - For circular-beam lamps only, an average luminous intensity distribution curve in the plane of the beam axis, in candelas, plotted on rectilinear coordinates
 - For non-circular-beam lamps, a polar intensity traverse through the beam axis, and a sufficient number of additional traverses (or planes) to define the beam pattern; where the distribution is reasonably uniform and detailed information is not required, a polar traverse and one horizontal traverse through the axis may be satisfactory
 - An isocandela diagram, for either circular-beam or non-circular-beam lamps

INFORMATIVE REFERENCES

- 1 American National Standards Institute. ANSI C78.379-2006, American National Standard - For Electric Lamps - Classification of the Beam Patterns of Reflector Lamps. Washington, DC: ANSI; 2006.
- 2 International Commission on Illumination (CIE). CIE 69-1987, Methods of Characterizing Illuminance Meters and Luminance Meters: Performance, Characteristics and Specifications. Vienna: CIE; 1987.
- 3 Illuminating Engineering Society. ANSI/IES LM-11-20, Approved Method: Photometric Testing of Searchlights. New York: IES; 2020.
- 4 Illuminating Engineering Society. ANSI/IES LS-4-20, Lighting Science: Measurement of Light – The Science of Photometry. New York: IES; 2020.
- 5 Illuminating Engineering Society. ANSI/IES LM-10-20, Approved Method: Photometric Testing of Outdoor Fluorescent Luminaires. New York: IES; 2020.
- 6 Illuminating Engineering Society. ANSI/IES TM-30-18, IES Method for Evaluating Light Source Color Rendition. New York: IES; 2018.
- 7 Illuminating Engineering Society. IES LM-15-03, IES Guide for Reporting General Lighting Equipment Engineering Data for Indoor Luminaires. New York: IES; 2003.

Process for Change to an ANSI/IES Standard under Continuous Maintenance

This standard is maintained under continuous maintenance procedures, for which IES has an established and documented program for regular publication of addenda or revisions, including procedures for timely, documented, consensus action on requests for change to any part of the standard. Committee consideration will be given to proposed changes by June 30 of any given year for proposed changes received by the IES Director of Standards no later than December 31 of the previous year.

Submittal Format

Proposed changes must be submitted to the IES Director of Standards in the announced published format. However, changes may be accepted in an earlier published format, if the differences are immaterial to the proposed change submittal. If the Director of Standards concludes that a current form must be utilized, the proposer may be given up to 20 additional days to resubmit the proposed changes in the current format.

Specific changes in the text or values are required and must be substantiated. Any change proposals that do not meet these requirements will be returned to the proposer. Supplemental background documents to support changes submitted may be included.

Submission to the Committee Chair

The Director of Standards shall forward proposed changes received on appropriate forms to the committee chair for assigning to committee members (responders) to develop responses to submitters of proposed changes.

Review and Clarification

Responders shall review proposals and should contact the proposer if necessary for clarification.

Response Recommendation

Designated responders shall draft a recommended committee response, including any recommended changes to the standard. The 'responders' recommended responses shall be submitted to the committee chair in electronic form usable by Society Staff, including any recommended change to the standard in response to proposals received.

Options for Committee response are limited to:

- a) Proposed change accepted for public review without modification
- b) Proposed change accepted for public review with modification
- c) Proposed change accepted for further study
- d) Proposed change rejected

The responders shall provide reasons for any recommendation other than option (a) above.

The designated responders shall not recommend option (c) unless the further study can be completed by October 1 of that year, and providing the Committee can then vote for option (a), (b), or (d) no later than November 15 of that year.

Editing

The Committee chair or his or her designee shall edit the draft responses and circulate the edited drafts to the committee for review.

Form for Proposing Change to an ANSI/IES Standard under Continuous Maintenance

NOTE: Use a separate form for each comment. Submit to the Director of Standards, IES, 120 Wall Street, 17th Floor, New York, NY 10005-4001. Email: standards@ies.org. Fax: 212-248-5017.

1. Submitter: _____
Affiliation: _____
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I hereby grant the Illuminating Engineering Society (IES) the non-exclusive royalty rights, including non-exclusive rights in copyright, in my proposals. I understand that I acquire no rights in publication of the standard in which my proposals in this, or other analogous, form are used. I hereby attest that I have the authority and am empowered to grant this copyright release.

Submitter's signature: _____ Date: _____

2. Title of publications and year published _____

3. Clause (section), sub-clause or paragraph number; and page number: _____

4. My proposal (check one):

- ☐ Change to read as follows
- ☐ Delete and substitute as follows
- ☐ Add new text as follows
- ☐ Delete without substitution

Use underscore to show material to be added (added) and strikethrough for material to be deleted (~~deleted~~). Use additional pages if needed.

5. Proposed change:

6. Reason and substantiation:

Select as applicable:

- ☐ Additional pages are attached. Number of additional pages: _____
- ☐ Attachments or referenced materials cited in this proposal accompany this proposed change.

Please verify that all attachments and references are relevant, current, and clearly labeled to avoid processing and review delays. Please list your attachments here:



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